



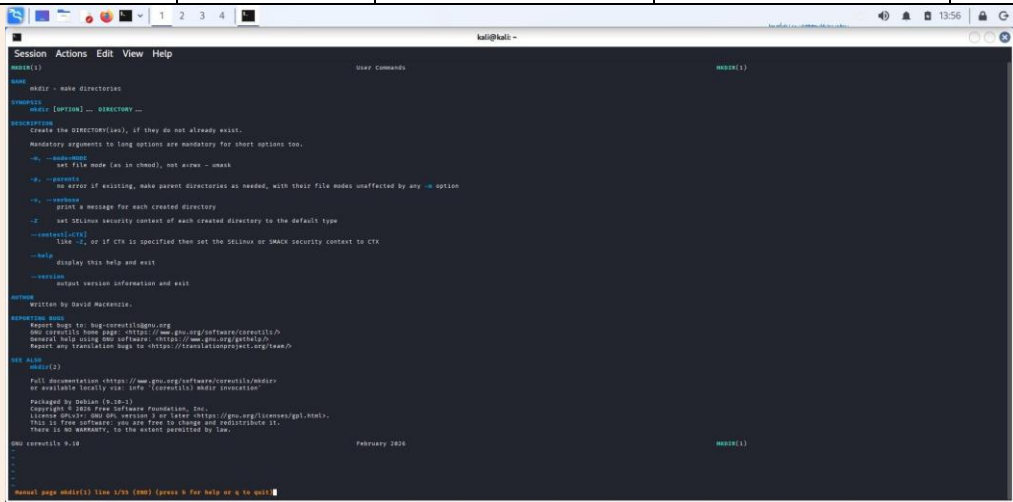
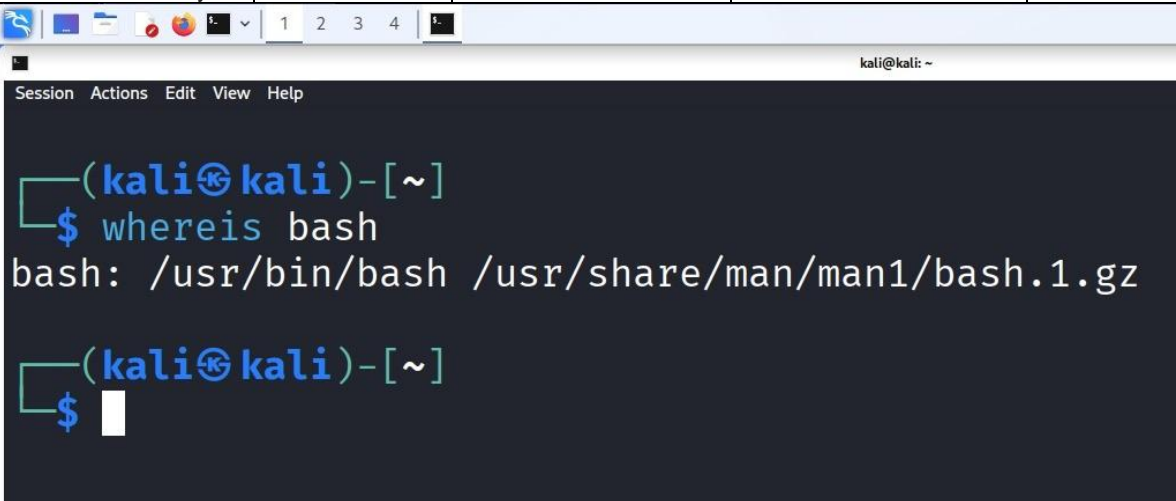
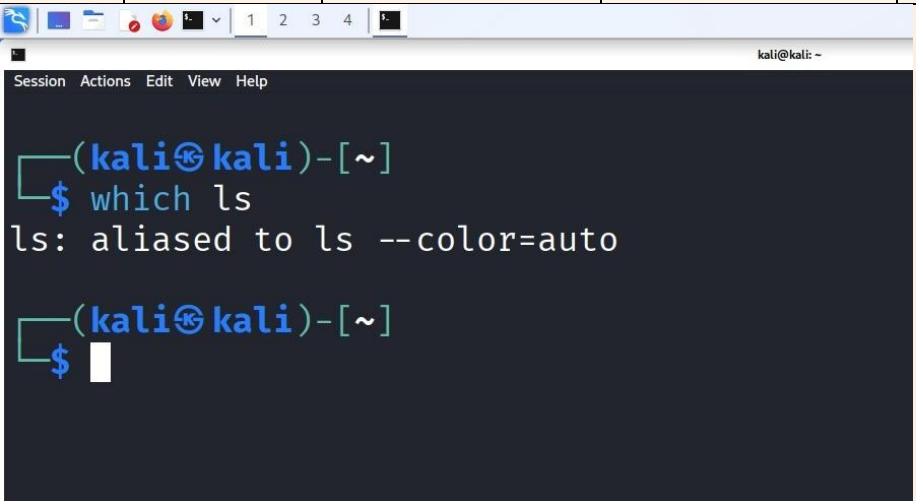
Manipal School of Information Sciences
Linux Operating System Lab

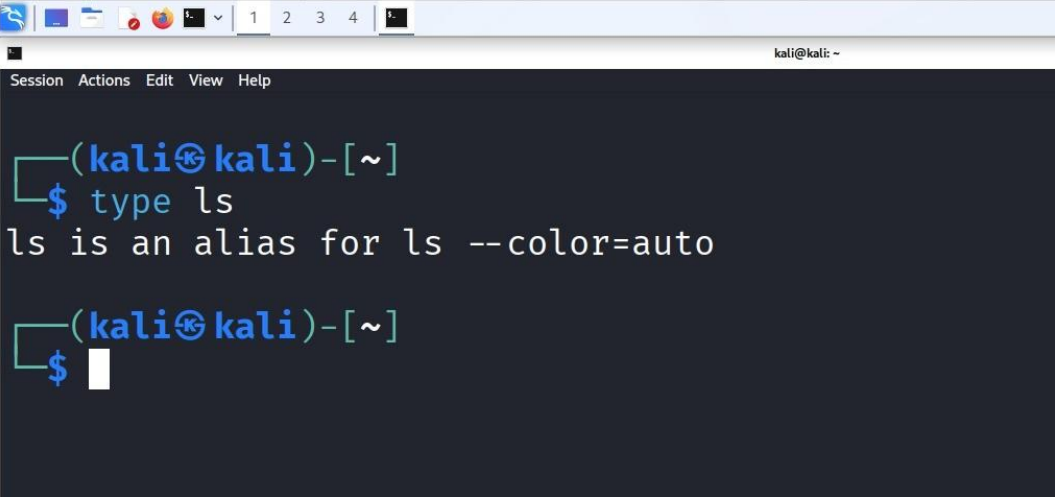
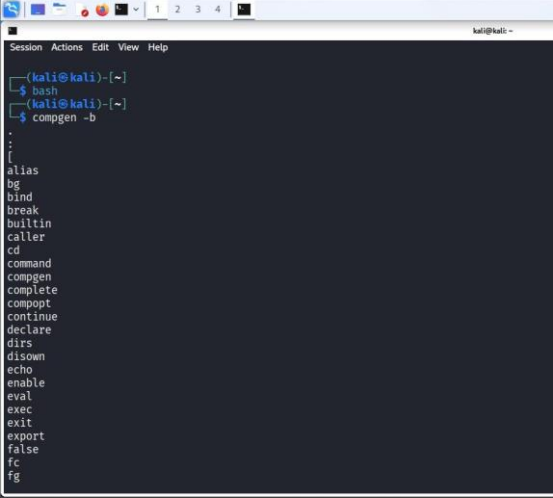
LAB ASSIGNMENT 02

Linux Laboratory Assignment 02

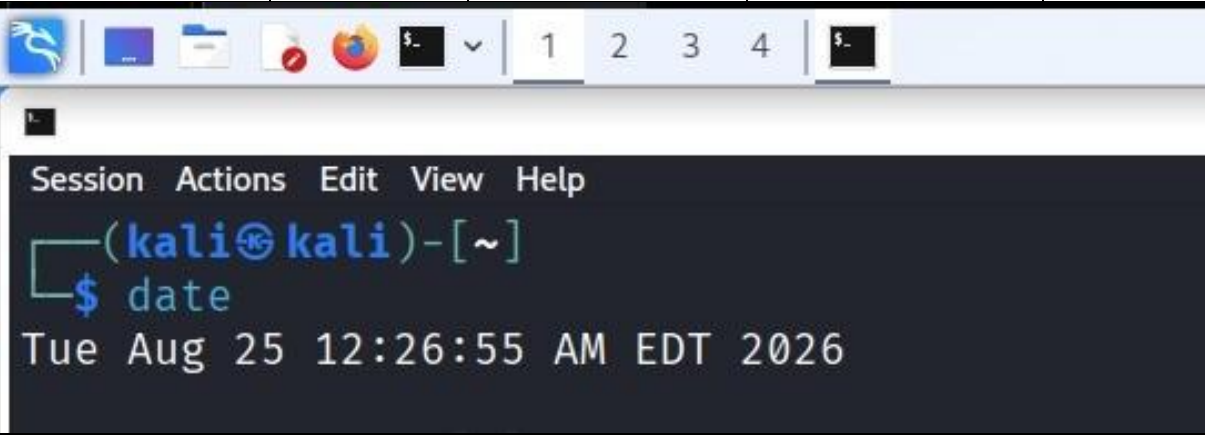
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Date:	25 August 2026
Subject:	Linux Operating System Lab

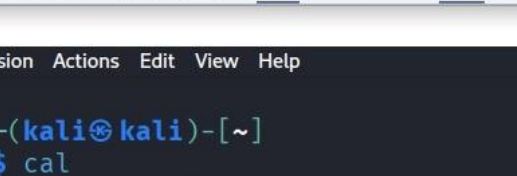
1. Getting Help

Task	Command Name	Syntax	Example	Screenshots
Manual page for a known command	man	man <command>	man mkdir	▼ below
				
Manual page for an unknown command	apropos	apropos	apropos copy	
Know the source file or binary	whereis	whereis <command>	whereis bash	▼ below
				
Know the path of a command	which	which <command>	which ls	▼ below
				
Determine whether command is external or internal	type	type<command>	type ls	▼ below

				
Help for an internal command	help	help <command>	help cd	
List Bash commands	compgen	compgen - c	compgen -b	▼ below
				
Know command usage	whatis	whatis <command>	whatis ls	

2. Basic Commands

Task	Command Name	Syntax	Example	Screenshots
Today's date	date	date	date	▼ below
				
Print calendar	cal	cal [month] [year]	cal	▼ below



The screenshot shows a Kali Linux terminal window. The window title bar includes icons for a web browser, file manager, and other applications, along with a tab labeled '1'. The terminal menu bar shows 'Session', 'Actions', 'Edit', 'View', and 'Help'. The prompt is '(kali㉿kali)-[~]'. The command '\$ cal' has been entered, and the output displays the calendar for August 2026. The days of the week are listed as Su, Mo, Tu, We, Th, Fr, Sa. The dates are arranged in a grid, with the first day of the month (1) falling on a Saturday.

```
(kali㉿kali)-[~]  
$ cal  
August 2026  
Su Mo Tu We Th Fr Sa  
1  
2 3 4 5 6 7 8  
9 10 11 12 13 14 15  
16 17 18 19 20 21 22  
23 24 25 26 27 28 29  
30 31
```

Print kernel version

uname

```
uname -r
```

```
uname -r
```

▼ *below*



The screenshot shows a Kali Linux terminal window. The top bar contains various application icons and a tab labeled '1'. The terminal's menu bar includes 'Session', 'Actions', 'Edit', 'View', and 'Help'. The prompt is '(kali@kali)-[~]'. The command 'uname -r' has been entered and executed, resulting in the output '6.19.14+kali-amd64'.

```
(kali@kali)-[~]  
$ uname -r  
6.19.14+kali-amd64
```

Print default shell

echo

```
echo $SHELL
```

```
echo $SHELL
```

▼ *below*

```
Session Actions Edit View Help

(kali㉿kali)-[~]
$ echo $SHELL
/usr/bin/zsh
```

Currently logged-in user

whoami

whoami

whoami

▼ *below*

Session Actions Edit View Help

```
(kali@kali)-[~]  
$ whoami  
kali
```

Create shortcut for command

alias

alias name='command'

alias list='ls -i'

▼ below

Session Actions Edit View Help

```
(kali@kali)-[~]  
$ alias list='ls -i'  
  
(kali@kali)-[~]  
$ list
```

```
786579 cryptography_lab2 786478 Documents 786479 Music 786480 Pictures 786477 Public 786481 Videos  
786474 Desktop 786475 Downloads 793481 permission_lab 786482 Projects 786476 Templates
```

Delete shortcut

unalias

unalias name

unalias list

▼ below

Session Actions Edit View Help

```
(kali@kali)-[~]  
$ unalias list
```

```
(kali@kali)-[~]  
$ list
```

```
Command 'list' not found, did you mean:  
command 'dist' from deb nmh  
command 'flist' from deb nmh  
command 'mlist' from deb mblaze  
command 'last' from deb wtmpdb  
command 'gist' from deb yorick  
command 'lift' from deb lift  
command 'hist' from deb loki  
Try: sudo apt install <deb name>
```

Change timestamp of file

touch

touch <file>

touch a.txt

▼ below

Session Actions Edit View Help

```
(kali@kali)-[~]  
$ touch a.txt
```

```
(kali@kali)-[~]  
$ ls -l a.txt
```

```
-rw-rw-r-- 1 kali kali 0 Aug 25 00:33 a.txt
```

```
(kali@kali)-[~]  
$ clear
```

Clear screen

touch

touch file1 file2

touch b.txt

▼ below

Session Actions Edit View Help

```
(kali@kali)-[~]  
$ touch b.txt
```

Create empty files

clear

clear

clear

▼ below

Session Actions Edit View Help

```
(kali@kali)-[~]  
$
```

Know disk usage

du

du [options] [path]

du -sh

▼ below

Session Actions Edit View Help

```
(kali@kali)-[~]  
$ du -sh  
406M .
```

Know free space

df

df [options]

df -h

▼ below

kali@kali: ~ Session Actions Edit View Help (kali@kali)-[~] \$ df -h Filesystem Size Used Avail Use% Mounted on udev 1.9G 0 1.9G 0% /dev tmpfs 389M 1.2M 387M 1% /run /dev/sda1 79G 17G 59G 22% / tmpfs 1.9G 4.0K 1.9G 1% /dev/shm none 1.0M 0 1.0M 0% /run/credentials/systemd-journald.service tmpfs 1.9G 8.0K 1.9G 1% /tmp none 1.0M 0 1.0M 0% /run/credentials/getty@tty1.service tmpfs 389M 108K 389M 1% /run/user/1000				
Linux release information	cat	cat /etc/os-release	cat /etc/os-release	▼ below
(kali@kali)-[~] \$ cat /etc/os-release PRETTY_NAME="Kali GNU/Linux Rolling" NAME="Kali GNU/Linux" VERSION_ID="2026.2" VERSION="2026.2" VERSION_CODENAME=kali-rolling ID=kali ID_LIKE=debian HOME_URL="https://www.kali.org/" SUPPORT_URL="https://forums.kali.org/" BUG_REPORT_URL="https://bugs.kali.org/" ANSI_COLOR="1;31"				
Show login history	last	last	last	▼ below
(kali@kali)-[~] \$ last kali tty7 :0 Tue Aug 25 00:26 - still logged in lightdm tty7 :0 Tue Aug 25 00:26 - 00:26 (00:00) kali tty7 :0 Mon Aug 24 13:39 - 14:23 (00:43) lightdm tty7 :0 Mon Aug 24 13:39 - 13:39 (00:00) kali tty7 :0 Mon Aug 24 09:08 - 09:28 (00:19) lightdm tty7 :0 Mon Aug 24 09:08 - 09:08 (00:00) kali tty7 :0 Fri Aug 21 04:34 - 06:30 (3+01:55) lightdm tty7 :0 Fri Aug 21 04:34 - 04:34 (00:00) kali tty7 :0 Fri Aug 21 04:10 - still logged in lightdm tty7 :0 Fri Aug 21 04:10 - 04:10 (00:00) kali tty7 :0 Wed Aug 19 01:44 - 01:46 (00:02) lightdm tty7 :0 Wed Aug 19 01:44 - 01:44 (00:00) kali tty7 :0 Tue Aug 18 23:53 - still logged in lightdm tty7 :0 Tue Aug 18 23:53 - 23:53 (00:00) kali tty7 :0 Tue Aug 18 09:31 - 09:34 (00:03) lightdm tty7 :0 Tue Aug 18 09:31 - 09:31 (00:00) kali tty7 :0 Tue Aug 18 09:23 - 09:30 (00:07) lightdm tty7 :0 Tue Aug 18 09:23 - 09:23 (00:00) postgres Tue Jun 16 10:29 - 10:29 (00:00) wtmpdb begins Tue Jun 16 10:29:31 2026				
Check current user privileges	sudo	sudo -l	sudo -l	
Find Linux release	lsb_release	lsb_release -a	lsb_release -a	▼ below

```

(kali㉿kali)-[~]
$ lsb_release -a
No LSB modules are available.
Distributor ID: Kali
Description:    Kali GNU/Linux Rolling
Release:       2026.2
Codename:      kali-rolling

```

Check system uptime

uptime

uptime

uptime

▼ below

```

(kali㉿kali)-[~]
$ uptime
00:40:21 up 14 min,  1 user,  load average: 0.08, 0.11, 0.09

```

Shutdown

shutdown

sudo shutdown -h now

sudo shutdown now

▼ below

```

Session Actions Edit View Help
(kali㉿kali)-[~/Desktop]
$ sudo shutdown now

```

Restart

reboot

sudo reboot

sudo reboot

▼ below

```

Session Actions Edit View Help
(kali㉿kali)-[~/Desktop]
$ sudo reboot

```

3. Navigation

Task	Command	Syntax	Example	Screenshots
Navigate to home directory	cd	cd ~	cd	▼ below


```
(kali㉿kali)-[~/permission_lab]  
$ cd
```

```
(kali㉿kali)-[~]  
$
```

Navigate to parent directory

cd

cd ..

cd ..

▼ below

```
(kali㉿kali)-[~]  
$ cd ..
```

```
(kali㉿kali)-[/home]  
$
```

Navigate to child directory

cd

cd <directory>

cd permission_lab

▼ below

```
(kali㉿kali)-[~]  
$ cd permission_lab
```

```
(kali㉿kali)-[~/permission_lab]  
$
```

Alternate command to cd

pushd

pushd <directory>

pushd permission_lab

▼ below

```
(kali㉿kali)-[~]
$ pushd permission_lab
~/permission_lab ~

(kali㉿kali)-[~/permission_lab]
$
```

Go back to previous directory

cd -

cd -

cd -

Go to root directory

cd

cd /

cd /

▼ below

```
(kali㉿kali)-[~]
$ cd /

(kali㉿kali)-[/]
$
```

4. File System

Task	Syntax	Command / Example
Identify the file system	df -T [path]	df -T .

5. Globbing

Create the practice directory and sample files with the following commands:

```
mkdir -p ~/glob_practice && cd ~/glob_practice
```

```
touch index.html login.html admin.htm report.txt access.log error.log user1.txt user2.log file123 backup1.html backup2.txt
```

Task	Pattern	Command
a. List all .html files	*.html	ls *.html
b. List all files other than .html files	!(*.html)	ls !(*.html)
c. Names containing at least one digit	*[0-9]*	ls *[0-9]*
d. user followed by a single digit	user[0-9].*	ls user[0-9].*
e. Ending in .txt or .log	+(*.txt *.log)	ls +(*.txt *.log)
f. Not ending in .txt or .log	*.{txt,log}	ls *.{txt,log}
g. Beginning with access or error	{access*,error*}	ls {access*,error*}
h. .html files containing a digit	*[0-9]*.html	ls *[0-9]*.html

6. Extended Globbing

Task	Pattern	Command
a1. Select access.log, error.log and auth.log	@(access.log error.log auth.log)	ls @(access.log error.log auth.log)
a2. Select all files except notes.txt	!(@(notes.txt))	ls !(@(notes.txt))
a3. Select auth.log.1 or auth.log.2	auth.log.@(1 2)	ls auth.log.@(1 2)
b. malware .log names containing a digit	malware*[0-9]*.log	ls malware*[0-9]*.log
c1. .log files containing failed_login or successful_login	@(failed_login successful_login).log	ls @(failed_login successful_login).log
c2. All .log files except firewall.log	!(@(firewall.log))	ls !(@(firewall.log))

7. Filters

Use the supplied Lab2_log_files directory for these exercises. Verified archive contents: login.log (50 lines), access.log (80 lines), security.log (120 lines), and incident.log (500 lines).

Item	Command
a. First 10 entries of login.log	head -n 10 login.log
b. First 5 entries of access.log	head -n 5 access.log
c. First 15 security events	head -n 15 security.log
d. First 20 incident events	head -n 20 incident.log
e. Only first entry in login.log	head -n 1 login.log
f. First 25 web requests	head -n 25 access.log
g. Last 10 login entries	tail -n 10 login.log
h. Last 5 access entries	tail -n 5 access.log
i. Last 20 security events	tail -n 20 security.log
j. Last 25 incident events	tail -n 25 incident.log
k. Only last security entry	tail -n 1 security.log
l. Last 50 incident events	tail -n 50 incident.log
m. login.log from line 41	tail -n +41 login.log
n. security.log from line 101	tail -n +101 security.log
o. Total lines in login.log	wc -l login.log
p. Total lines in access.log	wc -l access.log
q. Total security events	wc -l security.log
r. Total incident events	wc -l incident.log
s. Number of words in login.log	wc -w login.log
t. Number of bytes in security.log	wc -c security.log
u. Line, word, byte counts of access.log	wc -l -w -c access.log
v. Line counts of all four log files	wc -l *.log

Item	Command
w. First 20, then last 5	head -n 20 login.log tail -n 5
x. Last 20, then first 5	tail -n 20 login.log head -n 5
y. First 30 access, then last 10	head -n 30 access.log tail -n 10
z. Last 30 access, then first 10	tail -n 30 access.log head -n 10
aa. First 50 security, then last 10	head -n 50 security.log tail -n 10
bb. Last 50 security, then first 10	tail -n 50 security.log head -n 10
cc. login lines 11–20	sed -n "11,20p" login.log
dd. login lines 21–30	sed -n "21,30p" login.log
ee. access lines 11–20	sed -n "11,20p" access.log
ff. access lines 31–40	sed -n "31,40p" access.log
gg. security lines 21–30	sed -n "21,30p" security.log
hh. security lines 51–60	sed -n "51,60p" security.log
ii. incident lines 101–110	sed -n "101,110p" incident.log
jj. incident lines 201–220	sed -n "201,220p" incident.log
kk. incident lines 491–500	sed -n "491,500p" incident.log
ll. Count first 20 login entries	head -n 20 login.log wc -l
mm. Count last 15 access entries	tail -n 15 access.log wc -l

nn. Count words in first 50 security entries	head -n 50 security.log wc -w
oo. Count lines in last 25 security entries	tail -n 25 security.log wc -l
pp. Count extracted login lines 21–30	sed -n "21,30p" login.log wc -l
qq. Count extracted access lines 51–60	sed -n "51,60p" access.log wc -l
rr. Count incident events 101–150	sed -n "101,150p" incident.log wc -l
ss. First 100 incident, last 20, count	head -n 100 incident.log tail -n 20 wc -l